GROKBUILD HANDOFF PACK

Desk — Dual-Venue Trading Control Plane

NinjaTrader 8 ATI + Webull OpenAPI · paper-first · no strategy in v1

Date: 2026-09-03 | Repo: `desk` | Audience: implementer / operator peer

Classification: local-shareable spec. Secrets live off-repo. This file contains none.

Not investment advice. Not a signal service. No expected return is claimed. This package is a control plane and five operator desks. It does not pick trades. NinjaTrader simulation accounts cannot be copied to live. Webull sandbox fills are not live fills. Past paper results do not transfer. Automated trading remains the operator's trading. Venue terms of service apply.

1. Claim labels used throughout

Supported — checked against vendor docs on 2026-09-03. **Unproven** — not observed for this operator. **Disputed** — reject if anyone asserts it. Honest incomplete state is preferred over false readiness.

Item	State	Why
NT8 ATI file protocol (OIF + outgoing)	Supported	NinjaTrader help: Order Instruction Files + Information Update Files
Webull official Python OpenAPI	Supported	developer.webull.com 2026 docs + official SDK
Operator NT8 install / incoming path	Unproven	Not observed
Operator Webull API approval	Unproven	Review is 1–2 business days after apply
Capital, instruments, session hours	Unproven	Operator has not supplied them
Any strategy has positive expectancy	Unproven	No rules ship in v1
"This will print / AI trader / set and forget"	Disputed	Forbidden framing. Control plane only.

2. What ships vs what does not

You asked for a NinjaTrader + Webull trading bot, a spec an implementer can build, and five bots that operate it. What ships is the harness. What does not ship is an edge.

Ships in v1	Does not ship
Local Python app, repo name desk	Live trading enabled
Paper mode default; Sim* NT accounts only	Production Webull host as default
SENSE, RISK, ROUTER, BOOK, GOVERNOR	A starter EMA / RSI / VWAP playbook
NT ATI OIF writer (MOVE unique oif_*.txt)	Unofficial NT bridges or app scraping
Webull sandbox adapter + rate limiter	Options, crypto, event-contract adapters
Localhost dashboard + CLI kill	Cloud deploy, mobile app, trade-copier
Journal + recon + promotion checklist file	Guaranteed Sharpe, win-rate, or profit UI

3. Architecture

```
SENSE → GOVERNOR → RISK → ROUTER → NT8 ATI or Webull SDK
                            ↑              ■
                  BOOK ← acks / fills / positions ■■■■■■
```

GOVERNOR can halt any step. Only ROUTER may call a venue submit method.

Rail	v1 job	Official wire	Default
NinjaTrader 8	Futures execution	ATI OIF files + outgoing state files	Sim* account only
Webull	US stocks / ETFs	webull-openapi-python-sdk	api.sandbox.webull.com

NinjaTrader (Supported). Enable AT Interface, restart NT8. Write unique `oif*.txt` into `Documents\NinjaTrader 8\incoming`. MOVE, do not COPY. Same filename reuse or copy-in can file-lock and silently drop the order. ATI is not a market-data API and not a full brokerage API. PLACE must include a client ORDER ID or no outgoing order-state file is written. Outgoing folder is deleted on NT restart — treat that as Unproven book, not a flatten signal. If the Documents path is cloud-synced (OneDrive / iCloud / Dropbox), refuse the NT adapter.

PLACE format:
`PLACE;ACCOUNT;INSTRUMENT;ACTION;QTY;ORDERTYPE;[LIMIT];[STOP];TIF;[OCO];[ORDER ID];[STRATEGY];[STRATEGY ID]`

Other commands: CANCEL, CANCELALLORDERS, CHANGE, CLOSEPOSITION, CLOSESTRATEGY, FLATTENEVERYTHING, REVERSEPOSITION. CANCELALLORDERS and FLATTENEVERYTHING are **instance-global** and may hit accounts this desk does not own on that NT install.

Webull (Supported). Official SDK only. Do not hand-roll HMAC (vendor docs disagree SHA1 vs SHA256; SDK owns it). Sandbox first. Retail production keys need a brokerage account + application + about 1–2 business day review. Production 2FA token on first call. Reset-key invalidates the old key immediately. Rate limits that will bite: balances / positions / open-order detail **2 / 2s**; place / replace / cancel 600 / 60s; sandbox place 30 / 60s. US equities: market orders CORE session only; pre / after hours = limits only. v1 product scope is US stocks and ETFs. Options / crypto / futures adapters raise NotImplementedError.

4. Five operator desks

These are in-process modules, not five cloud AIs. They operate the desk. They do not invent edge. They cannot flip paper to live. Human final call is absolute.

Desk	Charter	Submit?	Veto?
SENSE	Watch an inbox (later, operator rules). Emit a Hypothesis, never a “signal.” Observation-only until a playbook exists.	No	No
RISK	Convert a hypothesis into a sized intent or a veto. Owns daily loss, qty, notional, cooldown, correlated book.	No	Yes
ROUTER	Translate one allowed intent into NT OIF or Webull preview→place. Thin adapters. No strategy logic.	Yes, gated	Malformed / unsupported only
BOOK	Official journal and recon. Venue is truth. Local ledger is a claim. Labels Supported / Unproven / Disputed.	No	Desync → T0 halt
GOVERNOR	Mode, session clock, localhost bind, secrets scan, human override, PDT / account fields.	No	Absolute

Runtime loop

- GOVERNOR issues a session token. Default mode = paper. Human ack required for any order.
- SENSE may emit a Hypothesis from the inbox. Invalid JSON is rejected, not traded.
- RISK returns ALLOW (qty may shrink, never grow), VETO, HOLD, or ESCALATE.
- If ALLOW and token valid and (not human-required or ack present) → ROUTER submits once. Idempotent client_order_id.
- BOOK consumes NT outgoing files and Webull events. Silent venue = Unproven, not filled.
- On daily-stop, disconnect, stale data, or recon break → T0 or higher per policy. Unlock is human-only.

5. Kill switch and flatten scope

Do not auto-flatten because outgoing/ is empty. NinjaTrader deletes that folder on restart. Empty files ≠ a flat account. First action on desync is halt new risk and alarm. Flatten is scoped to configured symbols unless a human asks for global.

Tier	Name	Effect	Who
T0	PAUSE	No new intent. Working orders stay.	RISK, BOOK, GOVERNOR, human
T1	CANCEL	Cancel working orders on configured symbols.	GOVERNOR / human; RISK may request
T2	SCOPED FLATTEN	NT CLOSEPOSITION per configured instrument+account. Webull cancel + offset.	GOVERNOR token; paper drills may auto
T3	GLOBAL	NT FLATTENEVERYTHING + Webull cancel/close all, then lock.	GOVERNOR + human dual-confirm only

Kill path is a separate module. No SENSE / model / strategy code on that path. Local panic file + CLI + localhost button. NT flatten still works if Webull is down (it is a filesystem write). Auto-resume after halt is forbidden.

6. P0 ship blockers (fail closed)

If a rule cannot be proven, do not ship that path. Implementer must encode these as first-class gates, not comments.

ID	Gate	Rule
P0-01	Live default off	MODE=paper. Live adapters not imported. No auto-promote from PnL or confidence.
P0-02	Live host load	Refuse start if Webull production host or non-Sim NT account is selected without arm.live.
P0-03	Secrets	No keys, tokens, or account numbers in git. Never log raw secrets. Mid-session 403 after key reset = HALT.
P0-04	Human final call	No order unless armed + GOVERNOR allow + RISK allow + unique client_order_id.
P0-05	Unspecified universe	Refuse size>0 if instruments, max qty, session, daily loss, or notional are still null.
P0-06	NT file contract	Temp then MOVE unique oif_*.txt. ORDER ID required. Cloud-synced Documents path = refuse NT.
P0-07	NT restart desync	outgoing/ wipe ≠ flatten. Mark book Unproven. Halt ROUTER. Rebuild. Do not PLACE.
P0-08	Flatten scope	Default = scoped CLOSEPOSITION. FLATTENEVERYTHING is T3, GOVERNOR + human only.
P0-09	Desync halt	Mismatch = T0 + alarm. Do not auto-T2 on first mismatch. Idempotent ids. No double-size on retry.

ID	Gate	Rule
P0-10	Stale / disconnect	Halt on stale quotes, NT DISCONNECTED, Webull fail, 429 burst. Token-bucket under venue limits.
P0-11	Session / product	No illegal session/order combo. Dual-venue correlated book (index futures vs index ETF) needs a cap.
P0-12	Kill drill	T0/T1/T2 drills on NT Sim* and Webull sandbox before arm.live can be created.
P0-13	OPSEC surface	Dashboard 127.0.0.1 only. Refuse 0.0.0.0. No personal identifiers in UI or shared artifacts.
P0-14	PDT / account facts	Collect cash/margin, equity band, PDT flag, options level before any Webull equity live path.
P0-15	Honest state	Do not invent fills. Do not claim edge. Promotion is a checklist file, not a vibe.

7. Regulatory facts (collect, do not assume)

Supported: a US margin account under $25,000 that places four or more day trades in five business days can be flagged Pattern Day Trader. Webull stocks and equity options can count. NinjaTrader futures do not use the FINRA PDT rule; they use futures margin and exchange day-margin rules. Webull options require an options-level approval. Index options are not on the retail OpenAPI surface locked here. A bot charter must not imply the venue blessed unattended live trading.

Unproven for this operator: cash vs margin, equity band, PDT flag, options level, day-trade count. Locked rule: if under $25k **or** flagged **or** unknown, RISK caps Webull equity/options day trades at 3 per rolling 5 sessions unless the operator documents a compliant policy. Wash-sale and tax lots are operator-owned and are not coded in v1.

8. What the operator still must supply

Until these exist, only paper-smoke is legal. The harness stays inert on purpose.

- Instrument list and NT front-month map (exchange format such as MNQ 09-26, not a continuous shortcut).
- Dollar caps in risk.yaml — shipped null so the operator must type them.
- Session hours actually intended (v1 ships a regular-hours stub only).
- Webull OpenAPI application, then sandbox keys in an OS secret store or untracked env. Never in this PDF, never in git, never in chat.
- Windows host with NT8, AT Interface enabled, NT restarted, Documents\NinjaTrader 8 not cloud-synced.
- Operator-authored rules. v1 ships an empty playbook stub. No default strategy.

9. Paper → live promotion (human checklist)

This is not a code path. Software never auto-flips the live flag. Stay paper if any row is Unproven.

#	Required evidence
1	Universe, size, hours, and loss caps written and unchanged for the paper window.
2	20 full sessions or 30 calendar days of complete journal (operator picks; default 20 sessions).
3	Reconcile match notes for mismatches; every break has a root-cause line.
4	T0 / T1 / T2 drills passed on NT Sim* and Webull sandbox in the last 7 days.
5	No P0 gate disabled.
6	Slippage and reject stats recorded. First live size = minimum increment.

#	Required evidence
7	Webull API approved. NT ATI enabled. Host is Windows. Documents folder is local.
8	Human written reason + two-confirm live flag. GOVERNOR holds the flag.
9	First live week: tighter daily loss cap; T0 after any unexpected reject.

10. Implementer build order

- Kernel + models + paper lock + append-only event bus.
- Paper adapter + tests (default mode is paper; live import refused).
- NT OIF atomic writer + outgoing poller against a Sim* account name.
- Webull sandbox client factory + rate limiter. No production host string in default config.
- RISK gates + scoped flatten. Separate kill module.
- BOOK journal + recon matcher. Desync = T0, not auto-T2.
- GOVERNOR CLI: status, arm, disarm, flatten, kill. Localhost dashboard.
- SENSE stub: file inbox only. Invalid files rejected.
- Promotion checklist file wired as data. Still blocked. Stop. Do not add a strategy.

11. Paste this to the builder

Build repo desk from this spec. Ignore any alternate-tree drafts. Do not invent a strategy. Do not enable live trading. Do not claim profit.

Goal: a local Python app that can run in paper/sim and (1) watch a hypothesis inbox, (2) gate every intent through GOVERNOR then RISK, (3) submit only via ROUTER, (4) talk to NinjaTrader 8 through official ATI OIF files (Sim* only in v1), (5) talk to Webull through official webull-openapi-python-sdk (sandbox host only in v1), (6) journal and reconcile in BOOK, (7) kill in T0 pause / T1 cancel / T2 scoped flatten.

Constraints: default mode=paper; no secrets in repo; NT writes unique file, temp then MOVE, always include ORDER ID; refuse NT adapter if path contains OneDrive / iCloud / Dropbox; Webull preview before place; official SDK only; FLATTENEVERYTHING is Governor dual-confirm only; dashboard 127.0.0.1 only; risk caps start null; no model in the kill path.

Done means: paper-smoke starts, prints health, writes one HEARTBEAT, places nothing, exits 0. Then stop.

12. Honest close

Most retail automation loses money. This desk exists so an operator can run rules they wrote, under gates they can read, with a kill switch that does not depend on the thing that is looping. It will not pick winners. If that is not acceptable, do not build it.

End of share pack. No personal identifiers. No secrets. No strategy. Repo name: desk.